

Written examination report for MC102 – Cyber Threat Intelligence
Kristiania University College
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Exam format: individual written home examination.

Final report format: recommended LaTeX or Word format and font 12 with 1.0 spacing. The limit is 25 pages max. A list of the bibliography/sources and other attachments can come in addition and attached to the report.

Grading scale: The Norwegian grading system uses the graded scale A - F, where A is the best grade, E is the lowest pass grade, and F is fail.

Weighting: 100% of the overall grade.

Support materials: All support materials are allowed.

Plagiarism control: We expect you to work independently. Please use citations/references and quotations if there is material you want to include in the report. If AI/LLMs were used, annotate where and for what reason.

General information:

- Start uploading your exam paper/file ahead of time, as it may take a long time to upload.
- Exam papers that are not handed in on Wiseflow by the specified time of the submission date will not be proceeded to assessment.
- Before submitting, remember to check that all files can be opened and that every file is included. It may be a good idea to check the saved files on several machines before submitting in Wiseflow.
- Incorrect file format or lacking documents may result in the submission not being passed or assessed.
- The total file size of the entire exam paper/file must not exceed 5GB in zip format or 5MB in PDF format.

Learning outcomes of MC102 - Cyber Threat Intelligence:

Knowledge. The student:

- has a good understanding of the Cyber Threat Intelligence (CTI) discipline
- has a good understanding of concepts, principles, and frameworks (e.g., diamond model, kill chain, ATT&CK) associated with CTI
- a clear understanding of the CTI cycle and its different applications as a support function
- sharing of CTI
- automation around CTI
- attribution

Skills. The student:

- can apply the threat intelligence cycle (establish a threat intel program) and analyze data and information in the context of deriving CTI
 - stakeholder management, direction, collection, processing, analysis, output, delivery, and feedback
- can identify, tailor, and utilize cyber threat intelligence frameworks and differentiate between strategic, operational, and tactical threat intelligence, in addition to delivering the right content in the right format as per the stakeholder needs/agreements
- can use threat data and threat intelligence to perform pivoting, develop detections, and use commonly utilized open-source tooling (e.g., scanning services, knowledge bases, CTI platforms, spreadsheets)
- will be able to operationalize threat intel in the context of a SOC environment and, in particular, SIEM/Incident Case Management/Threat Intelligence Platform, and exposure to API integrations

Exam Task

Establish a Cyber Threat Intelligence Program and Execute it

Task A - Stakeholder Management / Planning & Direction (30% of the grade)

1. Create a persona organization: An imaginary organization of your liking.
2. Establish and describe the organization's purpose (mandate)
3. Define and describe at least 2 CTI stakeholders of a different type (e.g., strategic, operational, tactical). They can be internal or external to your organization.
4. Identify and describe communication and feedback mechanisms pertaining to the stakeholder(s) and explain their importance.
 - a. Interaction with a stakeholder at any time during the process.
 - b. A way to retrieve feedback on our outputs after dissemination - determine the feedback/input you desire to have.
5. Identify and describe a use case that you will support as a CTI analyst. It can be one use case that will allow you to create two different deliverables, for example, one for the tactical audience and one for strategic (an abstract use case [that you need to tailor] could be related to relevant actively exploited vulnerabilities and the organization's security posture including risk and patch management).
6. Define Intelligence Requirements and, if necessary, Priority IR (at least 1). They need to be focused and precise.
7. Identify and describe the type of deliverable per the stakeholder requirements (e.g., textual/report, machine-readable - STIX2, TIP, CSV, spreadsheet)
8. Identify and describe how you will deliver your product/output to the stakeholders (medium for dissemination)
9. Identify and describe the periodicity of the deliverables
10. Explain how you will answer your IRs & PIRs and identify sources of information that will support answering your questions.
 - a. Create a Collection Management Plan
 - b. Identify and validate different types of sources and their reliability and accuracy (it is your first time preparing a CTI program and accessing various sources, so

you may need to determine reliability, accuracy, and redundancy during processing and analysis / or improvise). Elaborate on their importance and why they were chosen.

- c. Identify how you will collect information (e.g., internal intelligence, logs/telemetry, accessing external open source information, requests for information, Twitter feeds, third party, trusted groups, TIP, services such as Shodan/Malpedia/Abuse.ch/virus total, paid feeds, programmatically).
11. Identify frameworks that you will use for the overall process, including intelligence analysis frameworks (e.g., diamond model, ATT&CK framework, and other taxonomies) and tools (e.g., different software, including TIP if you need one)
12. If the deliverable is a report, create a style guide based on your audience (e.g., strategic, operational/tactical) and think of including infographics, timelines, etc. One figure equals one thousand words at times.

Hint: Intel471 material can help you a lot

Let's get hands-on and Initiate the Process

Task B - Collection (15% of the grade)

1. Collect the data/information/intelligence you need to answer your IRs/PIRs
 - a. Create a small table indicating what you collected to perform your analysis
 - i. You can include screenshots and highlight important information, for example, from “internal sources”, intelligence and threat reports and advisories, tweets, CVEs, ATT&CK info, etc. - you can save space by adding a URL of a shared folder with relevant sources so that we can, for example, see a folder structure that makes sense and allows you to be organized.
 - b. Take screenshots if you will use a threat intel platform or any kind of enricher.

Task C - Processing (15% of the grade)

1. *Process all the available data (in the way you deem necessary) and bring them into a format that can help your analysis (e.g., spreadsheets, TIP, Diamond model, kill chain). Be aware of redundancies and data errors. Spreadsheets are powerful and will save you time since you work individually, compared to installing special tools. The choice is yours.

*Processing involves organizing and structuring the information collected in the previous phase. It often requires removing duplicates and irrelevant information and converting it into a useful format for analysis. Group similar data together that can be used during the analysis phase.

Task D - Analysis & Production (30% of the grade)

1. Perform analysis on what you collected to derive intelligence.
2. Identify any gaps in the collection process and provide feedback so you can improve the collection process.
3. Bring everything together and account for cognitive biases if you aim to perform an assessment. If that is the case, describe a structured analytic technique that you would use to reduce cognitive biases. For example, elaborate on using “analysis of competing hypotheses” and consider alternatives.
4. Earlier, you processed and structured the information you needed; based on that assumption, describe/illustrate the associations you found to justify your assessment and product. It will help you in the preparation of your deliverable. You might deliver a strategic report, but it doesn't mean that you didn't perform tactical/operational analysis. Somebody at some point did. Get your facts together for your own benefit and, if needed, to support your deliverable.
 - a. Review, get feedback, fine-tune, and ensure high quality, accuracy, and actionability.
5. Prepare your output based on the format you have agreed with the stakeholders. Reflect on whether you have answered the intelligence requirement(s). - Prepare something concise; the report is up to 25 pages.

- a. Document the findings into a well-structured document (tactical vs operational vs strategic).
- b. Review, get feedback, fine-tune, and ensure high quality, contextuality, accuracy, and actionability.
- c. Add TLP and PAP markings if needed.

Task E - Dissemination (5% of the grade)

1. Briefly describe the process of disseminating your threat intel to the intended audience.
 - a. For example, was it a briefing? How long did it take to brief the organization? Maybe your output was a concise slide deck (your deliverable). - Everything is associated with what you have reported in Task A.
 - b. Provide your own feedback. Can you improve something?

Task F - Stakeholder Feedback (5% of the grade)

1. Assuming that YOU now are the stakeholder that received the threat intel product:
 - a. Submit your feedback based on the feedback request you received - you created a feedback process in Task A4.
2. Going back/reverting to your role as a CTI provider:
 - a. Briefly describe how you would act on the received feedback and its impact on the overall process, such as adjustments you need to perform in all stages of the CTI cycle.